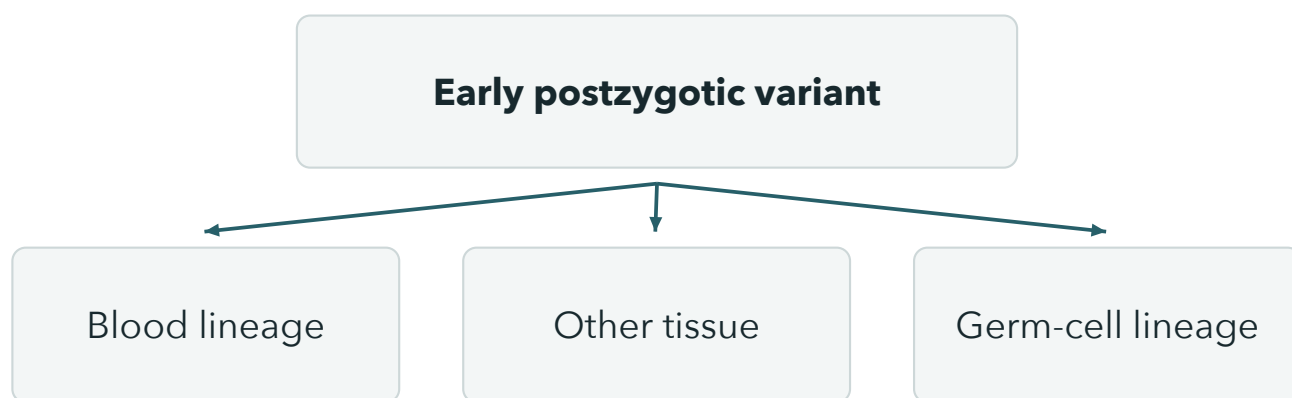


Tumour predisposition and mosaicism

Part 1 | A postzygotic change can vary across tissues

Blood is a sample of one lineage, not a complete census of every cell in the body.



Fractions may differ; blood does not sample every lineage

An illustrative early postzygotic variant is distributed unevenly among lineages; the proportions are hypothetical.

The central inference

A low-level RB1 variant can be missed by an assay whose detection threshold is above its blood allele fraction. A negative result reduces risk according to the test's performance but does not prove that no other tissue contains the variant. Use the assay limit rather than a universal negative-test rule.

Allele fraction is not the same as cell fraction

For an autosomal heterozygous change in a simple diploid mixture, variant reads sample only one of two alleles in affected cells.

Original diploid mixture

Variant-bearing cells	Expected allele fraction	Assumptions
20%	About 10%	One altered allele per affected cell
50%	About 25%	No copy-number imbalance
100%	About 50%	Constitutional heterozygosity

Model boundaries

Tumour copy-number changes, loss of heterozygosity, contamination and sequencing bias can alter these relationships. State the mixture assumptions before converting a measured allele fraction into an estimated proportion of cells.

Paired tissues can resolve the distribution

A tumour result and a non-neoplastic tissue result answer different questions.

Finding

Feature	What it supports	What it does not prove
Variant in tumour only	Somatic tumour alteration possible	Absence from all normal tissues
Same variant in normal tissue	Constitutional or mosaic distribution	Equal fraction in every lineage
Different normal-tissue fractions	Mosaicism plausible	Exact germ-cell fraction

Use independent samples

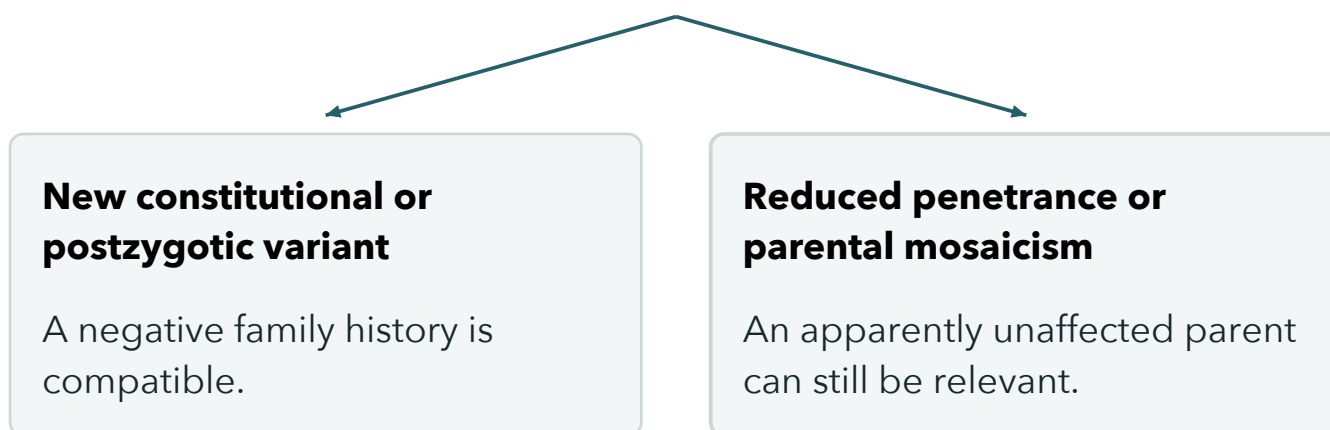
Deep sequencing of another non-neoplastic tissue can help distinguish a low-level mosaic pattern from a result confined to the sampled tumour. The additional specimen must be relevant and collected appropriately; another aliquot from the same original sample is not a second tissue.

Family history is incomplete biological evidence

A simplex presentation can arise by more than one route.

No affected relative is known

What remains possible?



Interpret cautiously

Do not equate unilateral presentation with zero heritable risk. Conversely, do not assign every patient the same recurrence estimate. Counselling integrates tumour presentation, molecular findings, assay sensitivity and family assessment through specialist genetics rather than one remembered percentage.

Testing is a probability update

A more sensitive assay can alter classification without changing the biological history.

An original testing sequence

- Conventional blood assay is negative
↓
- Clinical suspicion remains
↓
- Sensitive testing identifies a low allele fraction
↓
- Interpret tissue distribution and familial implications

Examination approach

Distinguish failure to detect from proof of absence. Then identify what a new result adds: analytical sensitivity, another tissue, or a familial comparison. These are separate ways of narrowing uncertainty and should not be described as equivalent repetitions of the same test.